Lab 11: Genomics & Biotechnology

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

Biotechnology puts our knowledge of DNA to work. Today, scientists can **cut** DNA at precise locations, **copy** a single molecule into billions, **sort** fragments by size, **identify** individuals by their unique DNA fingerprint, and even **edit** genes to treat diseases.

In this lab, you will explore the core tools of biotechnology. Think of these as tools in a scientist's toolbox!

Learning Objectives

1. Describe how scientists cut, copy, sort, and edit DNA.
2. Explain how the universal genetic code makes GMOs possible.
3. Apply DNA fingerprinting to forensic and inheritance scenarios.
4. Synthesize concepts from past modules (inheritance, replication) with modern biotechnology.

Part 1: The Molecular Toolkit

Before we dive into the details, let's look at the basic tools scientists use to manipulate DNA. Think of them by their action verb.

1. Match the scientific tool (A-E) to its everyday analogy and its action verb.
Write the correct letter in the blank space.

A. Restriction Enzyme
B. DNA Ligase
C. PCR (Polymerase Chain Reaction)

D. Gel Electrophoresis
E. CRISPR-Cas9

Letter	Everyday Analogy	Action Verb
	Molecular scissors	**CUTS** DNA at specific sequences
	Molecular glue	**GLUES** DNA pieces together
	Molecular photocopier	**COPIES** DNA millions of times
	Molecular size-sorter	**SORTS** DNA fragments by size
	"Find-and-replace"	**EDITS** specific DNA sequences

Part 2: PCR — Copying DNA in a Tube

PCR takes a tiny sample of DNA (from a crime scene, or a virus swab) and amplifies it. It works just like DNA replication in a cell, but controlled by a scientist changing the temperature!

2. Connect the steps of PCR in a test tube to the natural enzymes from DNA Replication in a cell (Module 07).

PCR Step (in a tube)	Temperature	What happens	Natural Enzyme Equivalent (in a cell)
Denaturation	~95°C	Heat separates the DNA strands	*(Hint: What unzips DNA?)*
Annealing	~55°C	Cool to let short	*(Hint: What adds the RNA primer?)*

PCR Step (in a tube)	Temperature	What happens	Natural Enzyme Equivalent (in a cell)
		primers attach	
Extension	~72°C	Warm to let a special enzyme build	______ *(Hint: What builds the new DNA strand?)*

3. DNA doubles with each PCR cycle. Fill in the math:
- Start: 1 copy
- After cycle 1: 2 copies
- After cycle 2: ______ copies
- After cycle 3: ______ copies
- After cycle 5: ______ copies
- After cycle 10: ______ copies *(Hint: 2^{10})*

Part 3: Gel Electrophoresis — Sorting

Gel electrophoresis uses electricity to pull DNA through a gel matrix.

4. DNA has a negative charge (from its phosphate backbone). Will it move toward the positive (+) or negative (-) electrode?

5. Imagine the gel is a thick forest of trees. If a giant bear and a small rabbit both try to run through the forest, the rabbit will get through faster. In a gel, which DNA fragments travel FARTHER — small pieces or large pieces?

Part 4: DNA Fingerprinting & Inheritance

You learned in Module 09 that you inherit half your alleles from mom and half from dad. The same goes for STRs (Short Tandem Repeats) — non-coding regions that we use for DNA fingerprinting.

6. Below is a mock DNA fingerprint for a paternity test. The child must inherit exactly half of their bands from the mother and half from the biological father.

Child's Bands: Position A, B, C, D
Mother's Bands: Position A, C

a) Which bands MUST have come from the father?

b) Suspect 1 has bands: B, D, E, F.
Suspect 2 has bands: A, B, G, H.
Who is the biological father?

Part 5: GMOs and The Universal Code

To make human insulin in a lab, we cut the human insulin gene out, and paste it into a bacterial plasmid (a small circle of DNA). The bacteria then read the human gene and build the insulin protein.

7. Step-by-Step GMO: Put these steps in order (number 1-4).

Order (1-4)	Step
	The bacteria read the gene and manufacture human insulin
	Use a **restriction enzyme** to cut the human insulin gene and the bacterial plasmid
	Put the recombinant plasmid into a bacterium
	Use **DNA ligase** to glue the human gene into the plasmid

8. The "So What?": Why is it possible for a bacterium to "read" a human gene and build the correct human protein? (Hint: Think back to Module 07—is the genetic code different for humans and bacteria?)

Part 6: Putting It All Together

Let's synthesize what you've learned across the last few modules!

9. A child is born with a genetic disorder caused by a single point mutation. The disorder follows an autosomal recessive inheritance pattern. Scientists want to explore using CRISPR to fix the broken gene.

Explain how concepts from Module 07 (DNA & Mutations), Module 09 (Inheritance), and Module 11 (CRISPR Biotech) all connect in this scenario. *(Write 3-4 clear sentences.)*